

```
distance = 0
for j in range(0, len(lines)):
    distance += math.sqrt(abs(lines[j][1][0] - lines[j][0][0])**2 + \
        abs(lines[j][1][1] - lines[j][0][1])**2)
return(distance)
```

This function takes a list of lines as its input and outputs the sum of the lengths of every line. Now that we have these functions, we can call them together with our itinerary to determine the total distance our salesman has to travel:

```
totaldistance = howfar(genlines(cities, itinerary))
print(totaldistance)
```

When I ran this code, I found that the `totaldistance` was about 16.81. You should get the same results if you use the same random seed. If you use a different seed or set of cities, your results will vary slightly.

To get a sense of what this result means, it will help to plot our itinerary. For that, we can create a `plotitinerary()` function:

```
import matplotlib.collections as mc
import matplotlib.pyplot as pl
def plotitinerary(cities, itin, plottitle, thename):
    lc = mc.LineCollection(genlines(cities, itin), linewidths=2)
    fig, ax = pl.subplots()
    ax.add_collection(lc)
    ax.autoscale()
    ax.margins(0.1)
    pl.scatter(x, y)
    pl.title(plottitle)
    pl.xlabel('X Coordinate')
    pl.ylabel('Y Coordinate')
    pl.savefig(str(thename) + '.png')
    pl.close()
```

The `plotitinerary()` function takes `cities`, `itin`, `plottitle`, and `thename` as arguments, where `cities` is our list of cities, `itin` is the itinerary we want to plot, `plottitle` is the title that will appear at the top of our plot, and `thename` is the name that we will give to the `png` plot output. The function uses the `pylab` module for plotting and `matplotlib`'s `collections` module to create a collection